Marc Kjerland, PhD

Applied ML Engineer and Quantitative Analyst

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(pronounced "chair"-land)

Summary

Applied machine learning (ML) engineer with 8+ years building, deploying, and independently validating models for financial crime detection, market forecasting, and risk in regulated environments. PhD in Applied Mathematics with global postdoctoral research.

Experience

Sep 2025 – Present **Bank of America**, *Quantitative Finance Analyst (VP)*, Global Financial Crimes, Chicago, IL

- Implemented anti-money laundering (AML) models impacting billions of transactions
- Built and shipped two AML detection models, cutting delivery time by 3-5x using AI coding agents
- Coordinated with development, technology, and business stakeholders to ensure quality deployment and compliance

Mar 2022 – Aug 2025 **KeyBank**, *Senior Quantitative Associate*, Model Risk Management, Remote

- Performed rigorous review of AI/ML & predictive models, primarily in fraud risk
- Validated dozens of models for regulatory compliance including performance monitoring, documentation, and code review
- Identified deficiencies in model soundness and performance drift, providing material feedback and challenger models
- Led junior validators, project management, and stakeholder communication

Jun 2019 – Mar 2022 **Bank of America**, *VP (Senior AI/ML Developer)*, Global Markets, Chicago, IL

- Developed & deployed Python ML framework for market forecasting and reporting
- Built reinforcement learning model for hedging application using custom TensorFlow/Keras neural network and compute layers
- Developed interpretable multi-label classifier to reduce human workload in securities compliance
- Derived evolving PCA method for robust covariance estimation from noisy market data

Feb 2018 – Jun 2019 **Verisk Analytics**, *Data Scientist*, Buffalo Grove, IL

- Led model development project with six data scientists, from data to deployment
- Built countrywide and state-specific insurance pricing models with data-driven recommendations for product owners
- Delivered technical presentations to stakeholders showing model improvement of 40–80%

2014 – 15, Fall 2017 **University of Illinois at Chicago**, *Postdoctoral Fellow*, Institute for Environmental Science and Policy, Chicago, IL

- Developed novel evaluation methodology for urban sustainability combining time series analysis and linear optimization

Jul 2015 – Jul 2017 **Kyoto University**, *Postdoctoral Researcher*, Disaster Prevention Research Institute, Kyoto, Japan

- Developed module for multiscale storm surge simulations using complex meteorological data
- Quantified hazard impacts of typhoons on Pacific coastlines using Monte Carlo simulation

Education

2015 **PhD, Applied Mathematics**, *University of Illinois at Chicago*
Thesis: Linear response closure approximations for multiscale systems

2005 **B.S., Mathematics**, *University of Minnesota, Twin Cities*

Skills

Programming: Python, SQL, SAS, C/C++, Fortran, JavaScript, Bash, LaTeX

Libraries: XGBoost, CatBoost, scikit-learn, statsmodels, TensorFlow, Keras

Platforms/Other: GCP, AWS, Teradata, Git, QGIS

Natural Languages: English, French, Japanese, German

Publications

2019 Sustainability assessment of universities as small-scale urban systems: a comparative analysis using Fisher information and data envelopment analysis. *Journal of Cleaner Production*

2017 Estimating climate change impacts on storm surge using adaptive mesh refinement. *Proceedings of Coastal Dynamics 2017*

2016 Impact assessment of climate change on coastal hazards in Japan. *Hydrological Research Letters*

2016 The response of reduced models of multiscale dynamics to small external perturbations. *Communications in Mathematical Sciences*